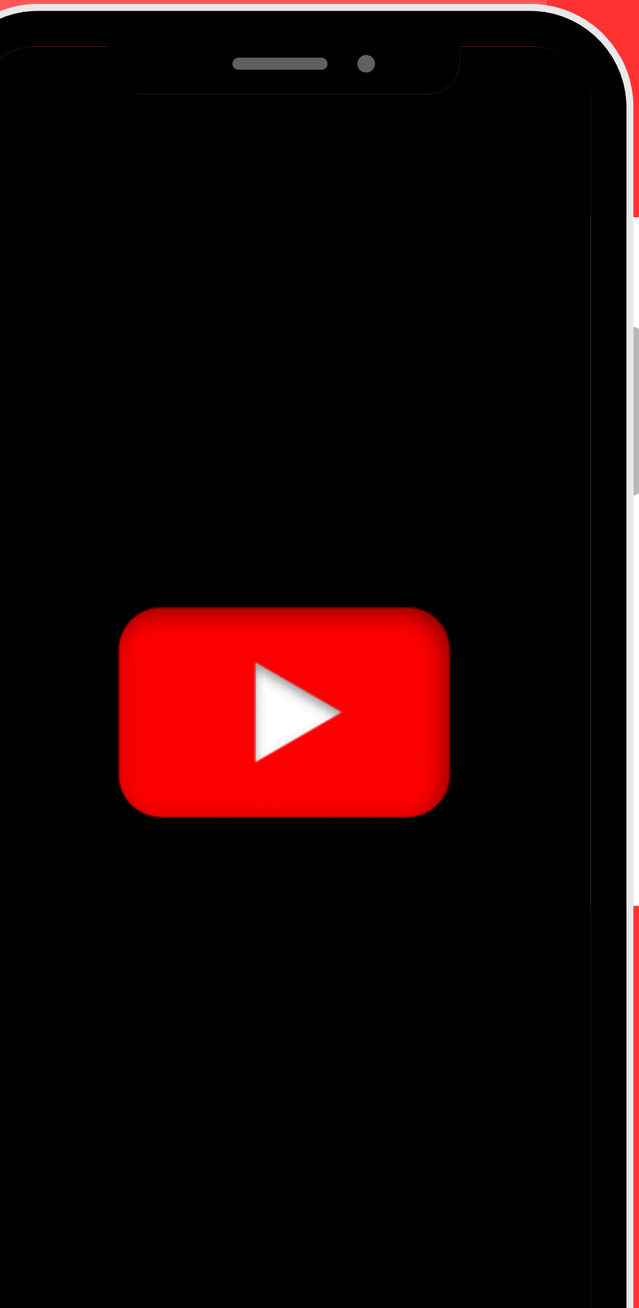


Youtube topic filter.

IT IS A CHROME EXTENSION THAT GIVES USERS CONTROL OVER WHAT CONTENT THEY SEE ON YOUTUBE.

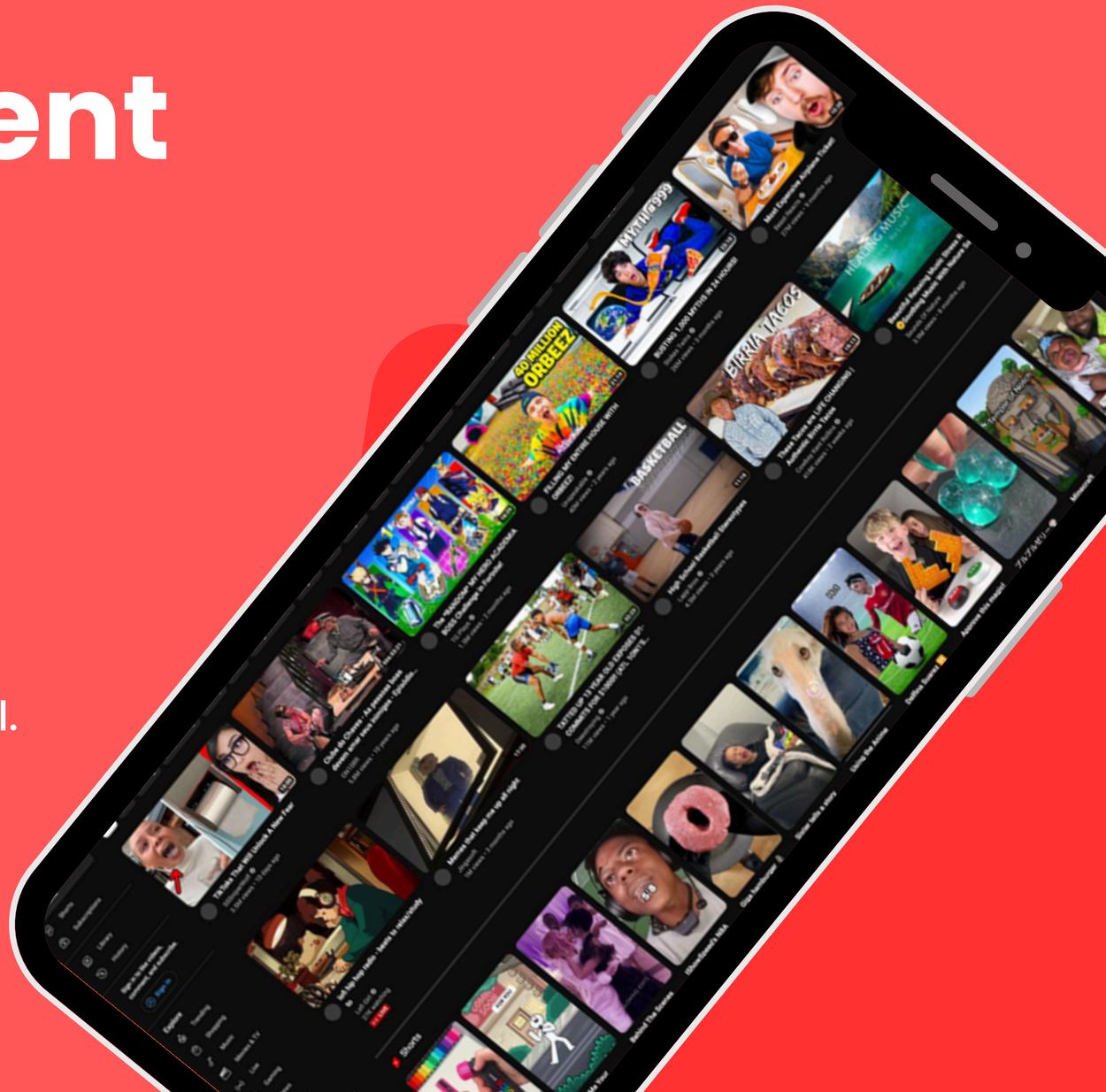
Problem Statement

"YouTube does not provide users with a built in filtering system."

Users cannot:

1) Block specific keywords
2) Filter entire topics
3) Avoid clickbait or sensational titles

This leads to distraction and lack of content control.
So we asked ourselves:
What if users could control their feed in real time?"



www.youtube.com

Input:
User enters keywords or selects mode

Processing:
Rule-based filtering + Topic detection + Sentiment scoring

Output:
Videos hidden or highlighted in real time

Second, our extension processes YouTube's video titles using:

- Keyword matching
- Topic detection
- Sentiment scoring

Challenges

Dynamic Loading:

Since YouTube loads content dynamically, we used a MutationObserver to ensure filtering updates in real time.

Lightweight NLP:

To keep the extension fast, we implemented ruleba sed topic mapping and heuristic sentiment scoring instead of heavy NLP models.

Shorts Rendering Challenge:

We also had a challenge filtering YouTube Shorts because they use a different layout, so we added Shorts renderer support to make filtering work there

Multiple Layouts:

We expanded our DOM selectors to support filtering across homepage, search results, sidebar, and Shorts

Over Filtering:

We introduced multiple filtering modes to give users control and avoid excessive content blocking